

Use case name :	Register a new Driver	
Scenario:	A new driver registers in the system.	
Triggering event:	Driver submits registration documents and details.	
Brief Description:	The driver provides required documents (license, car insurance, car model, type, color, and license plate number.), and the system admin reviews them for approval.	
Actors:	Driver	
Related use cases	need “check validation” to validate driver documents (validate document format , verifies authenticity and confirm phone number)	
Stakeholders:	Driver , system Administrator	
Preconditions:	• The driver has a valid driver’s license, car insurance, and a registered car. • Driver does not already have an active account.	
Postconditions:	If Approved: • Driver account is activated. • Driver is added to the system database. • Driver can log in and update availability status to appear in available drivers list. If Rejected: • Driver receives notification with reasons. • Driver can resubmit corrected documents.	
Flow of Activities:	Driver	System
	1. Submits driver’s license, car insurance, model, color, and plate number. 2. Receives approval/rejection notification. 3. Update availability states	1.1 Check validation 1.2 Stores data and notifies admin for review. If approved: 2.1 Activates account. 2.2 Add the new driver to data base of the system 2.3 Notifies driver 3.3 allows updates
Exception conditions:	1.1 Invalid Document Format(driver’s license, car insurance): System rejects upload and prompts for correct files (pdf). 1.1 Expired License/Insurance: Validation fails and driver must renew documents. 1.1 Duplicate Account: System detects existing account and blocks registration.	

Use case name :	Register a new Passenger	
Scenario:	A new passenger registers an account in the system.	
Triggering event:	Passenger enters phone number or email.	
Brief Description:	Passenger provides contact info then system sends verification code/email and upon successful verification, the account is activated.	
Actors:	Passenger	
Related use cases	"Update Passenger's Profile" and "check validation".	
Stakeholders:	Passenger	
Preconditions:	- Passenger has valid email or phone number. - Passenger does not already have an active account.	
Postconditions:	- Passenger is added to the system database. - Account is activated - passenger can log in and update his info	
Flow of Activities:	Passenger	System
	1. Enters email/phone. 2. Enters verification code. 3. Enters first and second name, phone number and payment details(optional) and email (optional)	1.1 validate email or phone number 1.2 Checks for duplicate account. 1.3 Sends verification code. 2.1. Verifies code matches sent code. 2.2. If correct: Ask for passenger info 3.1 check validation: - Activates account. - Adds passenger to DB. - notifies successful registration.
Exception conditions:	1.1 invalid email or phone number: User enters valid email or phone number. 1.2 Duplicate Account: System displays: "An account already exists with this email/phone." ; passenger enters new or not registered email or phone number 1.3 verification code sending Failure: System notifies passenger: "Failed to send code. Please try again later." 2.1 Verification Code Issues: Code Expired: System allows resend (max 3 attempts). Wrong Code: passenger enters code again and System blocks after 3 failed attempts (temporarily locks account). 3.1 Invalid Payment Info: Card number fails validation → system displays "Please check card details"; passenger try to enter card number again Expired card → "This card has expired"; passenger enters another card	

Use case name :	Update Passenger's Profile.	
Scenario:	Passenger updates their personal or payment information in the system.	
Triggering event:	Passenger requests profile update.	
Brief Description:	Passenger modifies name, payment method, etc.; system confirms changes.	
Actors:	Passenger.	
Related use cases	"Pay the Ride" (uses updated payment info), "Register Passenger" (prerequisite for having a profile), "check validation" for validate payment method updates.	
Stakeholders:	Passenger	
Preconditions:	• Passenger has an account. • Passenger must log in.	
Postconditions:	• Profile changes saved to database • Profile is updated	
Flow of Activities:	passenger	System
	1.chooses edit profile 2.Edits fields 3.Confirms update. 4.receives confirmation	1.1 Displays current profile info 2.1 Validates format. 3.1Verifies all entered data 3.2 Updates database 3.3 Sends confirmation
Exception conditions:	3.1 Invalid Payment Info: Card number fails validation → system displays "Please check card details"; passenger try to enter card number again Expired card → "This card has expired"; passenger enters another card 3.1invalid email or phone number: User enters valid email or phone number.	

Use case name :	Inquire a Trip	
Scenario:	Passenger checks available rides.	
Triggering event:	Passenger enters pickup/destination.	
Brief Description:	User chooses source and destination locations and System shows nearby cars with fare estimates and details.	
Actors:	Passenger	
Related use cases	"Register a new Passenger" Preceding use case. "Request a Car" Subsequent use case. "Calculate Fare" for calculating estimated fare based on car type ,trip distance and other factors . "Track Driver Location"	
Stakeholders:	Passengers, Drivers.	
Preconditions:	• Passenger is logged in • GPS/location enabled. • Available cars exist in nearby area.	
Postconditions:	List of available cars displayed along with car type, condition, availability of air conditioning, driver rate, and an estimated initial fare.	
Flow of Activities:	passenger	System
	1. Enables GPS or choose manual option to set pickup location. 2. Sets destination locations.	1.1 validate source locations: Valid locations. Checks service area coverage. 1.2 Queries driver database(Check availability) 2.1 validate destination locations: Valid locations. Checks service area coverage. 2.2 Estimates intial fares for each car type and add it to validate car lists. 2.3 Display cars available and car characteristics and intial fares. 2.4 Track drivers' location and display by the nearest and cheapest.
Exception conditions:	1.1 Invalid source address→alert message "invalid address reenter or use GPS " 1.2 driver's GPS is disabled so system can't track his location →notifies driver to turn on GPS when being available. 2.1 Invalid destination address→alert message "invalid address reenter the destination location". 2.2 driver's state changed →update displayed available list 2.2 No cars available → System suggests retry later or try another source location.	

Use case name :	Request a Car	
Scenario:	A passenger requests a ride after selecting a suitable car from available options.	
Triggering event:	The passenger selects a car and confirms the ride request.	
Brief Description:	The passenger chooses a preferred car from the list of available options, and the system sends the request to nearby drivers. Once a driver accepts, the passenger receives driver details and trip information.	
Actors:	Passenger	
Related use cases	• Inquire a Trip (Preceding use case) • Get available driver → handles requests by send request by the nearest available driver with highest rate and handles multiple drivers' request acceptance at the same time.	
Stakeholders:	Passengers ,Drivers	
Preconditions:	• The passenger is logged in. • Passenger has inquired the trip	
Postconditions:	• A ride request is sent to drivers. • The passenger sees driver details upon acceptance. • The trip is initiated when the driver arrives and trip state updated (ongoing). • Update driver availability (busy) • Track the ongoing trip (current location , estimated time to arrive , time passed since driver's acceptance)	
Flow of Activities:	passenger	System
	1. Selects a car and fare estimate. 2. Waits for driver acceptance. 3. Receives driver details 4. (Optional) Contacts driver via call/message.	1.1 Sends ride request to available drivers. 2.1 Notifies passenger when a driver accepts or cancels drivers' request. 2.2 in case of acceptance system updates driver availability to busy 3.1 Views driver details (name, rating, car info, ETA). 3.2 Updates real-time driver location on the map. 4.1 use End-to-end encryption for security of the communication and trip.
Exception conditions:	2.1No acceptance in 2 min → expired request and system try again or auto cancels the trip request after 3 trial and notifies passenger. 2.2 Multiple drivers' acceptance→the system chooses one and cancels the other and notifies drivers.	

Use case name :	cancel the ride			
Scenario:	A user (driver or passenger) cancels an ongoing ride request			
Triggering event:	User sends cancelation request			
Brief Description:	User sends a cancelation request and System notifies the other party and logs cancellation.			
Actors:	Passenger, Driver			
Related use cases	" Take action on issue" System administrator changes the rate of users who cancels requests frequently. " Update Passenger's Profile" or "validate payment"			
Stakeholders:	Passengers, Drivers, System administrator.			
Preconditions:	Ride is accepted but driver does not arrive the source location.			
Postconditions:	- Ride is canceled - Penalty may apply. - Update drivers availability (available)			
Flow of Activities:	Passenger	Driver	System	
	1. Cancel Ride request. 2. Write cancellation reason(optional) 3. Confirm cancellation	1. Cancel Ride request. 2. Write cancellation reason(optional) 3. Confirm cancellation	1.1 Verifies cancellation eligibility: - Time since request <= 2mins and small distance has been covered by the driver (No fees) Time >2min and large distance has been covered by the driver (fees) - Check that trip has not started 1.2 calculate fees 1.3 View fees 3.1 system make logs and sends it to system administrator to be reviewed 3.2 Notifies other party and updates trip status (canceled) and driver state(available) 3.3 apply fees : Check card if available , debit fees In cache payment add fees in the next ride	1.1 Verifies cancellation eligibility: Check that trip has not started 3.1 system make logs and sends it to system administrator to be reviewed 3.2 Notifies other party and updates trip status (canceled) 3.3 Update driver state(available)
Exception conditions:	3.3 no payment info for handle fees → system asks user to enter payment info 3.3 if passenger refuses payment ,system administrator suspend his account			

Use case name :	Pay the Ride	
Scenario:	Passenger pays after trip completion.	
Triggering event:	Ride completed and arrive destination location	
Brief Description:	Passenger selects cash/card, system processes payment. In case of cache, driver confirms payment. In case of credit update driver balance.	
Actors:	passenger	
Related use cases	"Validate payment method" and "Update Passenger's Profile."	
Stakeholders:	Passenger, driver.	
Preconditions:	In case of credit card payment method: • Passenger has valid bank account and credit card • Driver has valid bank account • Credit services are available	
Postconditions:	• The passenger's bank account will be debited • The Driver's bank account will be updated • Payment confirmation • trip state changes to (paid)	
Flow of Activities:	passenger	System
	1. Passenger send payment request. 2. Selects payment method (card/cash). 3.confirm payment	1.1Calulates final fare 1.2Ask for payment method 2.1 Processes payment (check there is valid card on the account) or prompts driver to confirm cash. 3.1 update trip as paid
Exception conditions:	2.1 Invalid Payment Info: Card number fails validation → system displays "Please check card details"; passenger try to enter card number again Expired card → "This card has expired"; passenger enters another card. If not solved system asks passenger to cache payment → if not available, driver and passenger Agreement on bank transfer to drivers account or to pay later using another method	

Use case name :	Rate The User		
Scenario:	Driver/passenger rates each other after the ride.		
Triggering event:	Users want to rate or complain.		
Brief Description:	Driver/passenger submits rating (1–5 stars) and give each other feedback.		
Actors:	Driver, passenger		
Related use cases	Take action on complaints		
Stakeholders:	Passengers, Drivers, system administrator		
Preconditions:	Ride is completed. Payment is completed		
Postconditions:	- Rating is saved; average score updated - Update user profile and database		
Flow of Activities:	passenger	Driver	System
	1. Selects rating and adds comments.	1. Selects rating and adds comments.	1.1 update database 1.2 Updates profile ratings and stores feedback. 1.3 notifies system administrator for reviewing
Exception conditions:	1.1 comments that contradicts policies → system rejects them and notifies user and system administrator		

Use case name :	Take action on issue	
Scenario:	Admin resolves disputes or deactivates accounts.	
Triggering event:	Complaint submitted or cancellation occurs.	
Brief Description:	Admin reviews issue and takes appropriate actions.	
Actors:	system administrator	
Related use cases	Validate Evidence: Admins verify chat logs/GPS data before acting.	
Stakeholders:	Admins, drivers, passengers.	
Preconditions:	- Complaint exists in the system. - Cancellation reasons - Logs of issues	
Postconditions:	- Action taken (e.g., warning, account deactivation). - Updates related to action taken	
Flow of Activities:	system administrator	System
	1.view complaints and logs Inbox. 2.Reviews complaint and logs. 3.Applies action (e.g., suspend user).	1.1 Provides evidence (ride logs, messages). 3.1 Notifies affected users. 3.2 updates (database, profiles)
Exception conditions:		

Use case name :	Generate report	
Scenario:	Admin requests daily system report.	
Triggering event:	Admin request daily report.	
Brief Description:	Admin requests daily report and System compiles ride stats, cancellations, ratings and returns it to the admin.	
Actors:	System administrator	
Related use cases	"Take action on issue"	
Stakeholders:	system administrator	
Preconditions:	System has daily ride data.	
Postconditions:	Report is generated. Actions are taken (if needed)	
Flow of Activities:	Admin	System
	1.Requests report. 2.Views/downloads report. 3.Take actions (if needed)	1.1 Compiles data (rides, cancellations, ratings). 1.2 Returns the compiled data to admin 3.1 Applies actions and notifies users
Exception conditions:	1.1 No data → System notifies admin.	